



Specification of the CA-500 Serial Interface on ASTM

NOTE : The standard specification is under discussion in NCCLS. This specification may be modified to match the standard when it will be published.

Ver. 1.01 α

SYSMEX CORPORATION

1. Scope

This document describes the conformity of the CA-500 to ASTM Serial Interface Standards.
(See NOTE)

[NOTE] ASTM (the America Society for Testing and Materials) A worldwide non-profit organization founded in 1898, with the purpose of establishing standards for specifications of materials, products and system services.

2. Overview

This document refers to the following two ASTM standards:

ASTM E 1381-91:

Specifications for Low-Level Protocols to Transfer Data Between Clinical Laboratory Instruments and Computer Systems.

ASTM E 1394-91:

Standard Specifications for Transferring Data Between Clinical Instruments and Computer Systems.

3. Communications Protocols

3.1 Hardware

3.1.1 Connectors

A DB-9-pin male I/O connector is located on the rear of the instrument.(See NOTE 1)

Table: Connector pin assignment

Pin No.	Signal name	Signal direction	Remarks
1	NC		Not used
2	Reception data RxD	IN	
3	Transmission data TxD	OUT	
4	Data terminal ready DTR	OUT	Not used in the ASTM
5	Signal ground SG		
6	Data set ready DSR	IN	Not used in the ASTM
7	Transmission request RTS	OUT	Not used in the ASTM
8	Transmission enabled CTS	IN	Not used in the ASTM
9	NC		Not Used

[NOTE 1] The ASTM standard specifies a DB-25-pin male connector as standard.

[NOTE 2] The ASTM specification does not include control signals.

For this reason, do not make connections to unused pins.

3.1.2 Signal identification level

Table : Signal identification level

Level	Data signal	Control signal
+3V or more	Logic "0", start bit	ON
-3V or less	Logic "1", stop bit	OFF

[NOTE 1] The ASTM does not use control signals.

3.1.3 Connection cable

The CA-500 uses a cable with a DB-9-pin female connector, in accordance with the following connection chart.

CA-500		HOST COMPUTER			
DB-9		DB-9		DB-25	
TxD	3	3	2	TxD	
RxD	2	2	3	RxD	
SG	5	5	7	SG	
RTS	7	7	4	RTS	
CTS	8	8	5	CTS	
DTR	4	4	20	DTR	
DSR	6	6	6	DSR	
NC	1				
NC	9				

3.1.4 Interface parameters

Setting interface parameters

Setting item	Parameter
Format	CA-500, CA-1000, <u>ASTM</u>
Baud rate	600, 1200, 2400, 4800, <u>9600</u> bps
Data length	7bit, <u>8bit</u>
Stop bit	<u>1bit</u> , 2bit
Parity	<u>None</u> , even, odd

The underlined parameters are the settings specified in the ASTM standard.

[NOTE] The ASTM standard allows for use of 7-bit data lengths, even or odd parity, and two stop bits, as required for special applications.

The ASTM standard does not allow for a baud rate of 600 bps.

The CA-500 does not relate to settings for 'Class' and 'Response Code(ACK/NAK)'.

3.1.5 Interface circuit

The CA-500 uses a SP232ACN microprocessor (MAX232) its interface circuitry.

3.1.6 Standard specifications (ASTM E 1381-91)

The physical layer of the CA-500 conforms to ASTM E 1381-91 "5. Physical Layers," except for connector type.

- (1) The CA-500 uses a DB-9-pin male connector.
(The ASTM standard specifies a 25-pin male connector.)

3.2 Transfer protocol

The data link layer transfers data between systems using a character-based protocol in accordance with ASTM E 1381-91 "6. Data Link Layers." This section describes communications control procedures. For details, refer to ASTM E 1381-91.

3.2.1 Communications modes

The data link layer involves the following three communications modes.

- (1) Setting modes :
Establishes a logical communications connection and determines the data transfer direction. In this mode, the sender and receiver are identified.
- (2) Transfer modes:
The sender transfers a message to the receiver.
- (3) Termination modes:
Breaks the communications connection, and both sender and receiver shift to idle mode.

3.2.2 Setting mode

- (1) The sender sends an [ENQ] signal to the receiver. To respond to the data from the sender, the receiver performs the following action:
 - Returns an [ACK] signal when communications are enabled.
 - Returns an [NAK] when communications are disabled.
 The sender waits for about 10 seconds before attempting to send an [ENQ] signal again.
- (2) When both sender and receiver send [ENQ] signals, the host computer passes control to the analyzer.
 - The analyzer attempts to resend an [ENQ] signal within one second after receiving the control signal from the host computer.
 - The host computer waits for 20 seconds before attempting to resend an [ENQ] signal.

3.2.3 Transfer mode

- (1) A block of text, as configured below, is to be transferred.

[STX] [F#] [text] [ETX] [CHK1] [CHK2] [CR] [LF]

If the volume of text exceeds 240 characters, it should be delimited with [ETB] signals as shown below, to split the text into two or more frames.

[STX] [F#] [text] [ETB] [CHK1] [CHK2] [CR] [LF]

- (2) [STX]: Start of text
- (3) [F#]: Frame number, from 0 to 7.
The frame number starts from 1, and is incremented by 1. That is, after 1 comes 2, 3, 4, 5, 6, 7, 0 and then 1 again.
- (4) For details on the [text] signal, see the relevant descriptions in ASTM E1394-91.
NOTE that the following ten types of transfer control characters, one type of format control character, and four types of instrument control characters are disallowed.
NULL (00), SOH (01), STX (02), ETX (03), EOT (04), ENQ (05), ACK (06), DLE (10), NAK (15), SYN (16), ETB (17), LF (0A), DC1 (11), DC2 (12), DC3 (13), DC4 (14)
(The values in parentheses are ASCII character codes.)
[NOTE] The ASTM standard makes no reference to NULL (00).
- (5) [ETX]: End of text
- (6) [CHK1] [CHK2]: Represented as a hexadecimal number, 0 to F.
Sums up the binary values corresponding to [STX] through [ETX]. This is then converted to a 2-digit hexadecimal value in the lower 8 bits of the ASCII code, representing 0 to F, and these digits are stored respectively as CHK1 and CHK2.
- (7) [CR] [LF]: End of frame

3.2.4 Termination modes

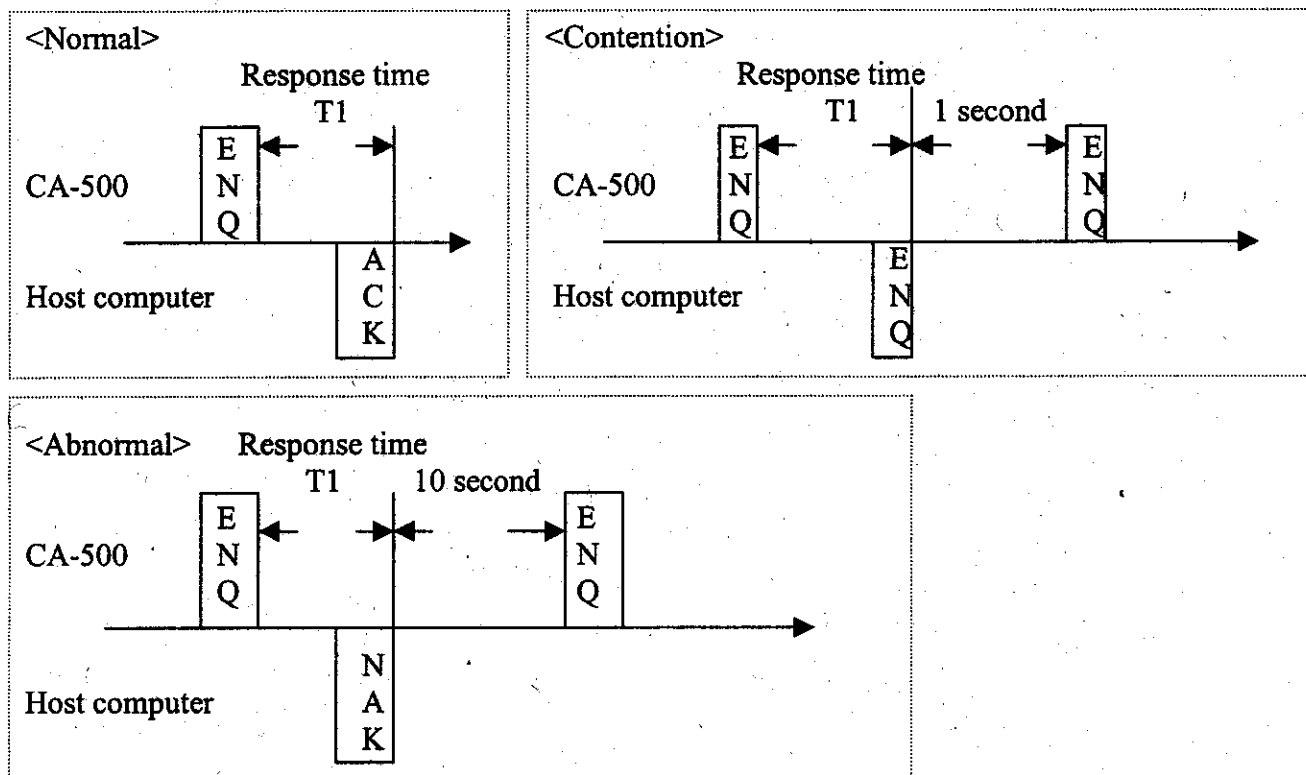
- (1) To notify the receiver that all the data has been transmitted, the sender sends an [EOT] signal and then shifts to idle mode.

3.2.5 Time-out

- (1) In setting mode, the sender sets the timer to 15 seconds when sending an [ENQ] signal. If the receiver does not reply within 15 seconds, the host computer assumes a time-out. If this occurs, the sender shifts to termination mode.
- (2) In transfer mode, the sender sets the timer to 15 seconds after sending the last character of the frame. If the receiver does not reply within 15 seconds, the host computer assumes a time-out. If this occurs, the sender shifts to termination mode.
The receiver sets the timer to 30 seconds when it first enters 'ready-to-receive' status, or it replies to the frame. If the receiver does not receive an [ACK] signal within 30 seconds, the host computer assumes a time-out. If this occurs, the receiver shifts to termination mode.

3.3 Communications procedure

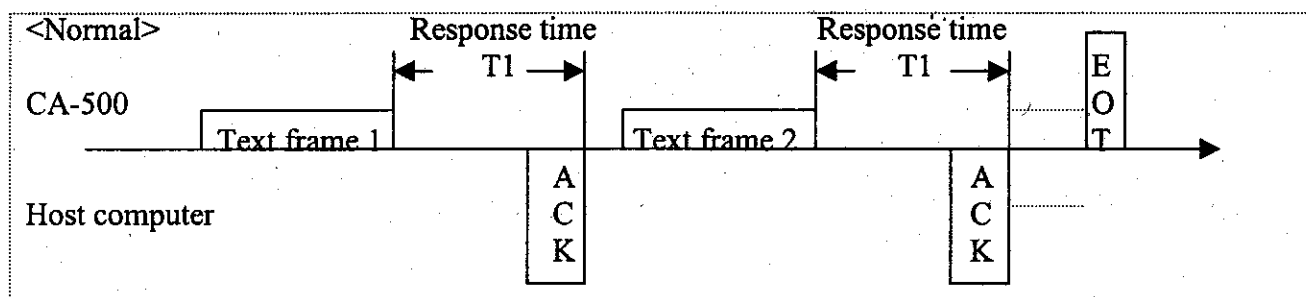
Setting mode



[Response time ($T1$) : $0.2\text{sec} < T1 < \text{Time-out time}$]

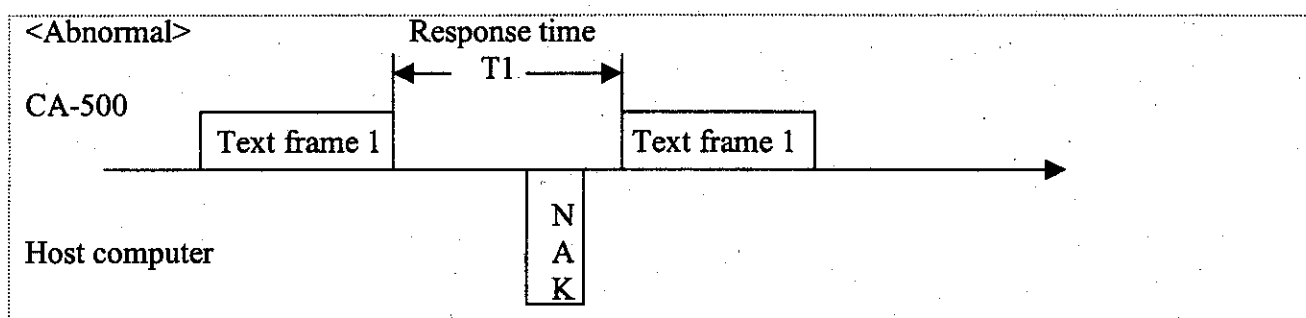
[NOTE 1] If the response time for the [ACK] or [NAK] exceeds 15 seconds, the CA-500 sends an [EOT] signal and returns to idle mode.

Transfer mode



[Response time (T1) : 0.2sec < T1 < Time-out time]

[NOTE 1] When the response time exceeds 15 seconds, the CA-500 sends an [EOT] signal and then returns to idle mode.



[Response time (T1) : 0.2sec < T1 < Time-out time]

[NOTE 1] When response time exceeds 15 seconds, the CA-500 sends an [EOT] signal and then returns to idle mode.

4. Text

4.1 Data and records

The CA-500 uses one of the formats of records specified in ASTM E 1394-91 for text (messages) to be transferred in frames in the form specified in ASTM E1381-91.

4.1.1 Records

A record is a series of text which starts with an ASCII character code, called a record identifier, and ends with a [CR].

Record type	Record identifier	Level	Description
Header	H	0	Contains the information on the messages from the sender and the receiver.
Patient information	P	1	Contains information about patient.
Request information	Q	1	Requests the test order information from the host computer
Test order	O	2	Contains information about test order.
Result	R	3	Contains information about test result.
Comment	C	1-4	Not used
Manufacturer information	M	1-4	Not used
Scientific information	S	N/A	Not used
Termination	L	0	Terminates the message.

- A high-level code includes the data of all the low-level records.
- A level number, except 0, must be placed after high-level records. However, information on the manufacturer (not used) and comment records can be inserted at any level in the layer and are considered to be one-level lower than the proceeding record.

4.1.2 Fields

A record is further divided into several fields by delimiters. A field is identified by its position within the record. The field length is variable. The following delimiters are used.

Delimiter type	Code	Description
Field		Separates the fields within the record.
Repeat	\	Separates the multiple sets of the same information within the field.
Component	^	Divides one field into several sub fields.
Escape	&	Used to embed a special character in the data.

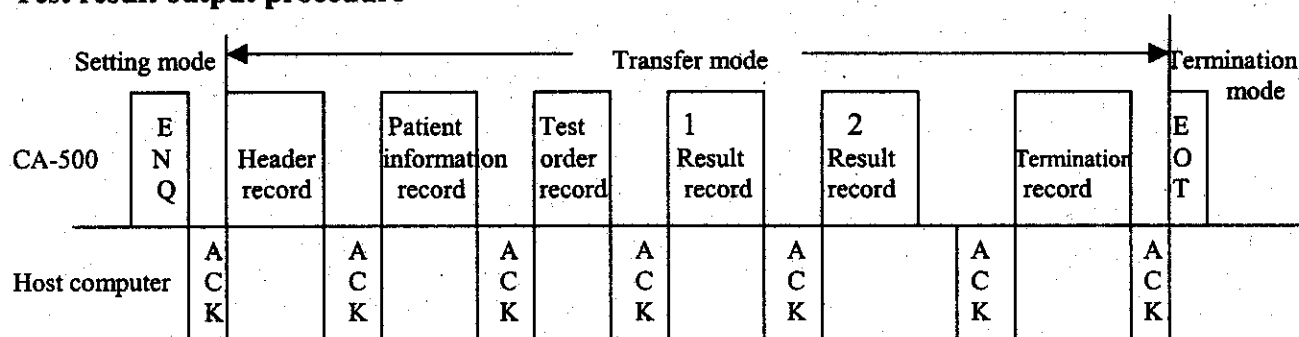
5. Communications from the CA-500

5.1 Test result output

The CA-500 outputs test results according to the following procedure

(1)	Reserve line	
(2)	Send header record	H \ ^ & CA-500 ^ SoftwareVersion i <CR>
(3)	Send patient information record	P 1 <CR>
(4)	Send test order record	O 1 RackNo. ^ TubePosition ^ SampleNo. ^ SampleNo.Attribute Priority ActionCode <C
(5)	Send test result record. (Repeat sending as many times as the number of results.)	R Sequence No. ^ ^ ^ TestParameterCode(table 5.1.1) ^ TestParameterName ^ DilutionRate ^ TestResultType Data Value Unit AbnormalFlag [TestCompletionDateAndTime <CR> R Sequence No. ^ ^ ^ TestParameterCode(table 5.1.1) ^ TestParameterName ^ DilutionRate ^ TestResultType Data Value Unit AbnormalFlag [TestCompletionDateAndTime <CR>
(6)	Send termination record	L 1 N <CR>
(7)	Release line	

Test result output procedure



5.1.1 Header record for test result output

H|\^&|||CA-500^SoftwareVersion|||||1<CR>

ASTM	Field name	Description	Remarks
7.1.1	Record Type	H	
7.1.2	Delimiter	, \, ^, &	Definition
7.1.3	Message Control ID		Not used
7.1.4	Access		Not used
7.1.5	Sender Name or ID	Instrument name^Software version^InstrumentNo.^InterfaceVersion^LogicalIDNumber	Instrument name : CA-500 Software version : Up to 6 characters Instrument No. : (CA-500 Not Used) Interface version : (CA-500 Not Used) Logical ID number : (CA-6000 Not Used)
7.1.6 - 7.1.12	Sender Street Address - Processing ID		Not used
7.1.13	Version No.		
7.1.14	Date and Time of Message		Not used

5.1.2 Patient information record for test result output

P|1<CR>

ASTM field	Field name	Description	Remarks
8.1.1	Record Type	P	
8.1.2	Sequence Number	1	
8.1.3 - 8.1.5	Practice Assigned Patient ID -Patient ID No. 3		Not used
8.1.6	Patient Name	Patient Name	Not used
8.1.7 -	Mother's Maiden Name - Dosage		Not used

5.1.3 Test order record for test result output

O | 1 | | Rack No. ^ Tube Position ^ Sample No. ^ Sample No. Attribute ^ | | Priority | | | | | Action Code <CR>

ASTM field	Field name	Description	Remarks
9.4.1	Record Type	O	
9.4.2	Sequence Number	1	
9.4.3	Specimen ID		Not used
9.4.4	Instrument Specimen ID	Rack No.^Tube position^ Sample No.^ Sample No. attribute^ Expanded order flag	Rack No.: 000000 - 999999 [NOTE : CA-500 displays only 2 characters from right.] Tube position: 00 - 99 Sample No.: Up to 15 characters Sample No. attribute : M (Manual input) A (Automatic allocation by the instrument) B (Read ID reader) C (Specify from the host computer) Expanded order flag: Not Used in the CA-500
9.4.5	Universal test ID		Not used
9.4.6	Priority	Priority	S (Stat Stat sample) R (Routine/general sample)
9.4.7 - 9.4.11	Requested/Ordered Date and Time - Collector ID		Not used
9.4.12	Action Code	Action Code	N (Normal patient test) Q (Quality control) S (Standard curve)
9.4.13 - 9.4.31	Danger code - Specimen Institution		Not used

[NOTE] The "Instrument Specimen ID"(5.1.3 ASTM field 9.4.3) of the test order record for test result output message is not same as the "Specimen ID"(5.2.6 ASTM field 9.4.4) in test order record for instrument input.

Example order : 000001^01^12345^M

Example results : 000001^01^12345^M^

5.1.4 Result record for test result output

R | Sequence No. | ^^^ Test Parameter Code ^ Test Parameter Name ^ Dilution Rate ^ Test result Type | Data Value | Unit | | Abnormal Flag | | | | | Test Completion Date and Time <CR>

ASTM	Field name	Description	Remarks
10.1.1	Record Type	R	
10.1.2	Sequence Number	Sequence No.	A serial number starting with 1
10.1.3	Universal Test ID	^^^Test parameter code(table 5.1.1)^ Test parameter name^Dilution rate^ Test result type	Test parameter code: Listed in the parameter code table. (table 5.1.1) Test parameter name: Up to 8 characters Dilution rate: Up to 3 characters (Example: 100% = 100) Test result type: 1 [Normal (real time)] 2 [Average (real time)] 5 [Normal (batch)] 6 [Average (batch)]
10.1.4	Data Value	Data Value	6 characters. For format, see the attached table table 5.1.2)
10.1.5	Unit	Unit	Up to 7 characters Although the ASTM standards define that the short form should be used, in accordance with ISO Standards, the CA-500 displays units in it
10.1.6	Reference Range		Not used
10.1.7	Result Abnormal Flags	Abnormal Flag	N (Normal test result) L (Below lower target limit) H (Above upper target limit) < (Below threshold lower limit) > (Above threshold upper limit) A (Analysis error)
10.1.8 - 10.1.12	Nature of Abnormality Testing - Date/Time Test Started		Not used
10.1.13	Date/Time Test	Test completion date and time	YYYYMMDDHHMMSS
10.1.14	Instrument Identification		Not used

[NOTE 1] Refer to 'Example test result output procedure'.(page 14/25)

[NOTE 2] The CA-500 may send the Data Value of result as "----." with flag "N", when the CA did not have the standard curve and without analysis error. This results are invalid and must be ignored.

[NOTE 3] The CA-500 always sends the PT-sec result additionally to %, INR,...

[NOTE 4] The CA-500 sends all measurement results in addition to the average result. If you want to proceed only the averages, ignore the results with Test Result Type 1, 5 of the Universal Test ID.

[NOTE 5] The Test Result Type "real time" is transferred, if the result is released by the auto-output. The Test Result Type "batch" is transferred, if the result is released by the hand out of the stored-data pool.

Table 5.1.1 : Parameter code for test result output

Parameter code	Parameter name	unit	Parameter code	Parameter name	unit	Parameter code	Parameter name	unit
041	PT	sec	181	VIII	sec	311	APL	dOD
042	PT %	%	182	VIII %	%	312	APL %	%
043	PT R	Ratio	191	IX	sec	321	Plg	dOD
044	PT INR	INR	192	IX %	%	322	Plg %	%
051	APTT	sec	201	X	sec	331	PC	dOD
061	Fbg	sec	202	X %	%	332	PC %	%
062	Fbg C.	mg/dL , g/L	211	XI	sec	511	TT	sec
121	II	sec	212	XI %	%	501	Fbg +	sec
122	II %	%	221	XII	sec	502	Fbg + C.	mg/dL , g/L
151	V	sec	222	XII %	%	521	Fbg -	sec
152	V %	%	301	AT3	dOD	522	Fbg - C.	mg/dL , g/L
171	VII	sec	302	AT3 %	%			
172	VII %	%						

045 der. Fib,

Table 5.1.2 : Data Value format

Data	Format
Clotting time ,Activity %	####.#
Ratio,INR	###.##
Conc.,dOD	##.###

[NOTE 1] '#' : (Space), '0'-'9', '*', '/', ', ',-'.

Example : '92.3', '****.*',

[NOTE 2] The decimal point gets transmitted .

[NOTE] Parameter name , parameter code, data format or unit name settings are able to change.
But please don't change these settings. This specification may be modified to match the standard NCCLS specification .

5.1.5 Termination record for test result output

L | 1 <CR>

ASTM field	Field name	Description	Remarks
13.1.1	Record type	L	
13.1.2	Sequence Number	1	
13.1.3	Termination Code	N	

Example test result output procedure

When the CA-500 reports the Re-dilution/Re-analysis and Reflex result, the CA-500 flags expand order flag. Expand order means that these test requests are ordered automatically by reflex function or re-analysis function or re-dilution function.

Single Measurement

The CA-500 reports test result as below.

Test Result

Double Measurement

Test Average Result

The CA-500 doesn't report the 1st result and the 2nd result in auto-output function, but report average result with flags. The 1st result and the 2nd are memorized in Stored Data function, and can report by Stored Data function.

Next page is example.

Single Measurement

Example : Result

```

CA-><ENQ>
HC-><ACK>
CA-><STX>1H|\^&||CA-500^00-01|1<CR><ETX><CHK1><CHK2><CR><LF>
HC-><ACK>
CA-><STX>2P|1<CR><ETX><CHK1><CHK2><CR><LF>
HC-><ACK>
CA-><STX>3O|1|001000^01^0000000000001^A^|R||||N<CR><ETX><CHK1><CHK2><CR><LF>
HC-><ACK>
CA-><STX>4R|1|^041^PT^100^1^^|
10.2|sec|N|||19970507182702<CR><ETX><CHK1><CHK2><CR><LF>
HC-><ACK>
CA-><STX>5R|2|^042^PT % ^100^1^^|
98.6|%|N|||19970507182702<CR><ETX><CHK1><CHK2><CR><LF>
HC-><ACK>
CA-><STX>6R|3|^043^PT R.^100^1^^|
0.89||N|||19970507182702<CR><ETX><CHK1><CHK2><CR><LF>
HC-><ACK>
CA-><STX>7R|4|^044^PT INR^100^1^^|
0.79||N|||19970507182702<CR><ETX><CHK1><CHK2><CR><LF>
HC-><ACK>
CA-><STX>0R|5|^051^APTT^100^1^^|
32.6|sec|N|||19970507182702<CR><ETX><CHK1><CHK2><CR><LF>
HC-><ACK>
CA-><STX>1L|1|N<CR><ETX><CHK1><CHK2><CR><LF>
HC-><ACK>
CA-><EOT>

```

Double Measurement

Example : Average result

```

CA-><ENQ>
HC-><ACK>
CA-><STX>1H|\^&||CA-500^ 00-01|1<CR><ETX><CHK1><CHK2><CR><LF>
HC-><ACK>
CA-><STX>2P|1<CR><ETX><CHK1><CHK2><CR><LF>
HC-><ACK>
CA-><STX>3O|1|001000^01^0000000000001^A^|R||||N<CR><ETX><CHK1><CHK2><CR><LF>
HC-><ACK>
CA-><STX>4R|1|^051^APTT^100^2|
32.6|sec|N|||19970507182702<CR><ETX><CHK1><CHK2><CR><LF>
HC-><ACK>
CA-><STX>5R|2|^061^Fbg^100^2|
8.2|sec|N|||19970507182702<CR><ETX><CHK1><CHK2><CR><LF>
HC-><ACK>
CA-><STX>6R|3|^062^Fbg C.^100^2|
220.0|mg/dL|N|||19970507182702<CR><ETX><CHK1><CHK2><CR><LF>
HC-><ACK>
CA-><STX>7L|1|N<CR><ETX><CHK1><CHK2><CR><LF>
HC-><ACK>
CA-><EOT>

```


5.2 Request and instruction of test order

Request instructions conform to the following procedure.

Output of request from the CA-500

(1)	Reserve line	
(2)	Send header record	H \ ^ & CA-500^Software version 1 <CR>
(3)	Send request information record	Q 1 Rack No.^Tube position^Sample No.^Sample No. attribute ^^Test parameter code(Table 5.2.1)^Test parameter name O Request start date and time<CR>
(4)	Send termination record	L 1 N <CR>
(5)	Release line	

Input of test instruction from the host computer

(6)	Reserve line	(Reserve the line from the host computer.)
(7)	Send header record	H \ ^ & Host Name^^^ CA-500<CR>
(8)	Send patient information record	P 1 <CR>
(9)	Send test order record	O 1 Rack No.^Tube position^Sample No.^Sample No. attribute ^^Test parameter code 1^^100^ \ ^^Test parameter code 2^^100^... Priority Request date and time Action code <CR>
(10)	Send termination record	L 1 N<CR>
(11)	Release line	

Request instruction flow

instruction	Setting mode	Request transfer mode			Request completion mode	Setting mode	Test instruction transfer mode			Test completion
CA-500	E N Q	Header record	Request record	Termination record	E O T	A C K	A C K	A C K	A C K	A C K
Host computer	A C K	A C K	A C K	A C K	E N Q	Header record	Patient information record	Test record	Termination record	E O T

5.2.1 Header record for request output

H|\^ & ||| CA-500^Software version ||||| 1 <CR>

ASTM field	Field name	Description	Remarks
7.1.1	Record Type	H	
7.1.2	Delimiter	, \, ^, &	Definition
7.1.3	Message Control ID		Not used
7.1.4	Access Password		Not used
7.1.5	Sender Name or ID	Instrument name^Software version^Instrument No.^Interface version^Logical ID number	Instrument name : CA-500 Software version : Up to 6 characters Instrument No. : (CA-500 Not Used) Interface version : (CA-500 Not Used) Logical ID number : (CA-500 Not Used)
7.1.6 - 7.1.12	Sender Street Address - Processing ID		Not used
7.1.13	Version No.		
7.1.14	Date and Time of Message		Not used

5.2.2 Request record for request output

Q | 1 | Rack No. ^ Tube position ^ Sample No. ^ Sample No. attribute || ^^ Test parameter code (Table 5.2.1) ^ Test parameter name | O | Request start date and time <CR>

ASTM field	Field name	Description	Remarks
12.1.1	Record Type	Q	
12.1.2	Sequence Number	1	
12.1.3	Starting Range ID Number	Rack No.^Tube position^ Sample No.^Sample No. attribute	Rack No.: 000000 - 99999 [NOTE : CA-500 displays only 2 characters from right .] Tube position: 00 - 99 Sample No.: Up to 15 characters Sample No. attribute : M (Manual input) A (Automatic allocation by the instrument) B (Read ID reader)
12.1.4	Ending Range ID Number		Not used
12.1.5	Universal Test ID	^^Test parameter code (Table 5.2.1)^Test parameter name	Test parameter code: See the attached sheet. (Table 5.2.1) Test parameter name: Up to 8 characters When specifying multiple parameters, use the repeat symbol as shown below: ^^Test parameter code1^Test parameter name1^^Test parameter code2^Test parameter name2 ^^Test parameter code3^Test parameter name3
12.1.6	Nature of Request Time Limits	0: Request date and time	The ASTM standard does not define specifications for date/time requests.
12.1.7	Beginning Request Results Date and Time		YYYYMMDDHHMMSS
12.1.8 - 12.1.13	Ending Request Results Date and Time - Request Information Status Code		Not used

Table 5.2.1 : Parameter code for request output

Parameter code	Parameter name	Parameter code	Parameter name	Parameter code	Parameter name
040	PT	180	VIII	300	AT3
050	APTT	190	IX	310	APL
060	Fbg	200	X	320	Plg
120	II	210	XI	330	PC
150	V	220	XII	500	Fbg +
170	VII	510	TT	520	Fbg -

[NOTE] Parameter name or parameter code settings are able to change.

But please don't change these settings.

This specification may be modified to match the standard NCCLS specification .

5.2.3 Termination record for request output

L | 1 | N <CR>

ASTM field	Field name	Description	Remarks
13.1.1	Record type	L	
13.1.2	Sequence Number	1	
13.1.3	Termination Code	N	

5.2.4 Header record for test instruction input

H|\^&|||Host Name ^^^^|||CA-500<CR>

ASTM field	Field name	Description	Remarks
7.1.1	Record Type	H	
7.1.2	Delimiter	\,^,&	Definition
7.1.3	Message Control ID		Not used
7.1.4	Access Password		Not used
7.1.5	Sender Name or ID	Host Name ^{^^}	Up to 16 characters. However, the current version does not check the name or ID of the sender.
7.1.6 - 7.1.9	Sender Street Address - Characteristics of Sender		Not used
7.1.10	Receiver ID	Instrument name ^{^^} Instrument No. ^{^^}	Instrument name : CA-500 Instrument No. : CA-500 Not Used
7.1.11 - 7.1.14	Comment or Special Instructions -Date and Time of Message		Not used

5.2.5 Patient information record for test instruction input

P|1<CR>

ASTM field	Field name	Description	Remarks
8.1.1	Record Type	P	
8.1.2	Sequence Number	1	
8.1.3 - 8.1.5	Practice Assigned Patient ID -Patient ID No. 3		Not used
8.1.6	Patient Name	Patient Name	Not used
8.1.7 -	Mother's Maiden Name - Dosage		Not used

5.2.6 Test order record for test instrument input

O | 1 | Rack No. ^Tube position ^Sample No. ^Sample No. attribute | | ^^Test parameter code
1^^100\ ^^Test parameter code2^^100... | Priority | Request date and time | | | | Action code<CR>

ASTM field	Field name	Description	Remarks
9.4.1	Record Type	O	
9.4.2	Sequence Number	1	
9.4.3	Specimen ID	Rack No.^Tube position^ Sample No.^Sample No. attribute	Rack NO., Tube position : Outputs values that match the requests. Sample No.: Up to 15 characters Sample No. attribute : M (Manual input) A (Automatic allocation by the instrument) B (Read ID reader) C (Specify from the host computer)
9.4.4	Instrument Specimen		Not used
9.4.5	Universal test ID	^^Test parameter code (Table 5.2.2)^^ Dilution rate^Option	Test parameter code: See the attached table. (table 5.2.2) Dilution rate: Up to 3 characters. For the format, see the attached table.(table 5.2.3) Option: Not Used in the CA-500 When requesting multiple parameters, use the repe symbol as below: ^^Test parameter code 1^^100\ ^^Test parameter cod 2^^100\ ^^Test parameter code 3^^100\ ^^Test parameter code 4^^100 ...
9.4.6	Priority	Priority	S (Stat, emergency patient), R (Routine/general patient)
9.4.7	Requested/Ordered Date and Time	Request date and time	YYYYMMDDHHMMSS
9.4.8 - 9.4.11	Specimen Collection - Collector ID		Not used
9.4.12	Action Code	Action code	N (Normal patient test) Q (Quality control) [Refer to NOTE 1]
9.4.13 - 9.4.31	Danger code - Specimen Institution		Not used

[NOTE 1] QC Sample No.

Please use 'QC01'-'QC12' only.

[NOTE 2] The "Specimen ID"(5.2.6 ASTM field 9.4.4) of the test order record for instrument input message is not same as the "Instrument Specimen ID"(5.1.3 ASTM field 9.4.3) in test order record for test result output.

Example order : 000001^01^12345^M

Example results : 000001^01^12345^M^

[NOTE 3] Do not use the Option of the "Universal Test ID" of the test order record for instrument input. The option can not be used in the CA-500.

Table 5.2.2 : Parameter code for test instrument input

Parameter code	Parameter name	Parameter code	Parameter name	Parameter code	Parameter name
040	PT	180	VIII	300	AT3
050	APTT	190	IX	310	APL
060	Fbg	200	X	320	Plg
120	II	210	XI	330	PC
150	V	220	XII	500	Fbg +
170	VII	510	TT	520	Fbg -

Table 5.2.3 : Dilution rate format

Dilution rate	Format
100%	100

[NOTE] Parameter code settings are able to change. But please don't change these settings. This specification may be modified to match the standard NCCLS specification .

5.2.7 Termination record for test instruction input

N<CR>

ASTM field	Field name	Description	Remarks
13.1.1	Record type	L	
13.1.2	Sequence Number	1	
13.1.3	Termination Code	N	

Example : Request instruction procedure

Request

```

CA-><ENQ>
HC-><ACK>
CA-><STX>1H|\^&|||CA-500^ 00-01|||||1<CR><ETX><CHK1><CHK2><CR><LF>
HC-><ACK>
CA-><STX>2Q|1|000009^01^          ^M||^040^PT\^050^APTT\^060^Fbg
\^120^
  II\^150^V\^170^VII\^200^X\^180^VIII\^190^IX\^210^XI\^220^XII
    \^300^AT3\^310^APL|O|19970506192956<CR><ETX><CHK1><CHK2><CR><LF>
HC-><ACK>
CA-><STX>3L|1|N<CR><ETX><CHK1><CHK2><CR><LF>
HC-><ACK>
CA-><EOT>

```

A host computer outputs indication immediately after inputting request .

```

HC-><ENQ>
CA-><ACK>
HC-><STX>1H|\^&|||HostName^^^^|||CA-500<CR><ETX><CHK1><CHK2><CR><LF>
CA-><ACK>
HC-><STX>2P|1<CR><ETX><CHK1><CHK2><CR><LF>
CA-><ACK>
HC-><STX>3O|1|000009^01^970506-001-01^C||^040^^100\^050^^100\^060^^100
|R|19970506193005|||N<CR><ETX><CHK1><CHK2><CR><LF>
CA-><ACK>
HC-><STX>4L|1|N<CR><ETX><CHK1><CHK2><CR><LF>
CA-><ACK>
HC-><EOT>

```

[NOTE] It is reeled and returned , as far as aforementioned procedure is even 10 from 1 of tube position every 1 rack .